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Fast and non-destructive determination of active pharmaceutical ingredient concentration in an electrospun nanofiber patch using infrared spectroscopy

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ABSTRACT

Infrared (IR) spectroscopy has been successfully utilized to non-destructively determine the concentration of α–arbutin (ABT) loaded on cellulose acetate (CA) electrospun-nanofiber patches. Initially, the homogeneity of the ABT distribution over the CA nanofiber mat, which would directly influence the sample representation of the acquired transmission IR spectra, was evaluated. IR microscopic mapping performed on the 3.0 wt% ABT sample covering an area of 800 × 800 μm^2 revealed that ABT concentrations were moderately homogeneous (relative standard deviation: 3.3%). Another issue to consider for accurate analysis is the potential thickness variation from sample to sample, which makes the ABT peak intensity considerably less concentration-representative. The absolute amount of loaded ABT varies depending on the thickness of the nanofibers, thereby causing the intensity of the ABT peak to fluctuate. To resolve this problem, a method involving the peak area ratio between the ABT and CA band was used. When the IR sampling area was 800 × 800 μm^2, the correlation between the ABT concentration (1.5–4.8 wt%) and the peak area ratio was linear, but the accuracy was not sufficient. Increasing the sampling area to 4 × 4 mm^2 yielded better accuracy with a standard error of prediction (SEP) of 0.10 wt%. The acquisition of more concentration-representative sample spectra with wide area coverage was attributed to accurate concentration determination. In addition, the potential of utilizing near-infrared (NIR) and Raman spectroscopy for the same analysis was evaluated and discussed.

1. Introduction

Recently, nanofibers possessing a high specific surface area and area-to-volume ratio have been widely utilized for numerous applications in energy generation/storage, water treatment, environment remediation, healthcare, and biomedical engineering due to their unique physicochemical properties and characteristics [1,2]. Since nanofibers are able to form networks of highly porous mesh with remarkable interconnectivity, their applications in biomedical and tissue engineering such as wound dressing, drug delivery [2,3], and nutraceutical delivery [4] have shown great promise. In particular, nanofiber-based drug delivery systems can facilitate accelerated dissolution [5–8] and increase the solubility of small molecules [9,10], making them useful in controlling drug delivery over time for local [11–17] or systemic drug administration [18–21].

To measure and control drug entrapment efficiency and/or drug loading efficiency, it is necessary to measure the active pharmaceutical ingredient (API) concentration in nanofibers. For this purpose, two different methods have been generally employed. The first is the exhaustive extraction or elution of API from nanofibers; for example, the flush of API loaded on ion exchange nanofibers using a solution with high salt concentration [14]. In this method, the complete extraction of API needs to be ensured. However, this approach is rather questionable in many instances since the incorporated API does not easily elute from a nanofiber matrix. To overcome the issue, the complete dissolution of a whole API-embedded nanofiber with suitable solvents has been alternatively adopted [6,22]. In this second method, complete extraction of API can be ensured. However, the concurrently present polymer often interferes with subsequent quantitative analysis of API such as using HPLC. Overall, both methods require moderate extraction or complete dissolution of a sample, which are laborious and time-consuming processes. Therefore, a non-destructive method able to determine the content of loaded API in nanofiber formulations with no sample pretreatment is highly demanded.

To achieve the goal, vibrational spectroscopy is a most desirable candidate since information on API loaded in nanofibers could be directly obtained without further sample pretreatment. As is known, due to the capability of fast and non-destructive analysis, its usage has

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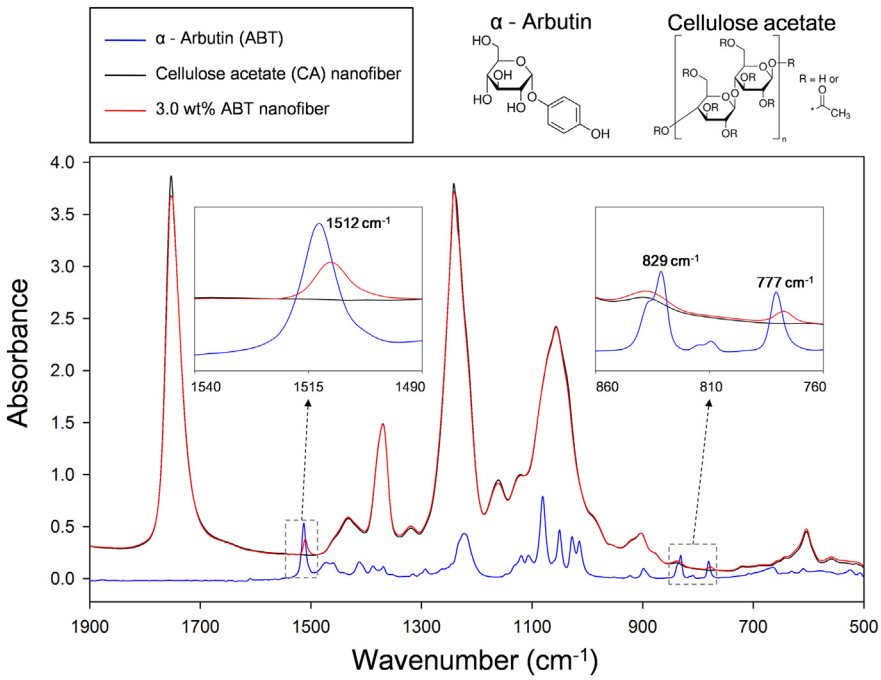

Fig. 1. IR spectra of pure ABT, pure CA nanofiber, and 3.0 wt% ABT nanofiber samples in the 1900–500 cm^{-1} range. The insets highlight the ABT peaks at 1512 cm^{-1} and 777 cm^{-1}. The molecular structures of ABT and CA are also shown. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

gradually increased for quality control of final pharmaceutical products [23–27] and process monitoring under a process analytical technology (PAT) framework [28]. Here, to evaluate the potential, α–arbutin (ABT) loaded cellulose acetate (CA) electrospun-nanofiber mats (simply refer as ABT/CA nanofibers) containing different ABT concentrations were prepared. The molecular structures of ABT and CA are shown in Fig. 1. ABT is a glycosylated hydroquinone that has an inhibitory function against tyrosinase and is a skin whitening agent used in creams, lotions, sera, foams, gels, mask sheets, soap bars, tablets, and capsules [29]. Initially, using infrared (IR) spectroscopy, IR mapping was performed over a 3.0 wt% ABT nanofiber sample to examine the microscopic distribution of ABT. If the distribution of ABT is inhomogeneous, the accuracy will degrade when a sampling area is not sufficiently large to handle the inhomogeneity. Next, IR spectra of ABT/CA nanofibers were acquired by covering sample areas of 800 × 800 μm^2 and 4 × 4 mm^2, and the accuracies in determining the ABT concentrations for both cases were compared. Since the thickness of nanofibers varies from sample to sample, the absolute amount of loaded ABT varies accordingly. To compensate for the unwanted erroneous fluctuation of peak intensities induced by the thickness variation, the peak ratio between ABT and CA bands was used to determine the concentration. Finally, near-infrared (NIR) and Raman spectra of the 3.0 wt% ABT sample were also collected, and both spectral features were examined to evaluate their utility for the same analysis.

2. Experimental section

2.1. Fabrication of electrospun ABT nanofibers

Cellulose acetate (CA; MW: 30 kDa, degree of acetylation: approx. 2.4), α-arbutin (ABT; purity ~98.0%), sodium metabisulfite (purity ~98.0%), and phosphomolybdic acid were purchased from Sigma–Aldrich. The solvents used in this study were of analytical grade. Distilled water was used throughout the experiments.

The CA nanofibers were prepared by an electrospinning method. Initially, a 17 wt% CA solution was prepared by dissolving 1.02 g of CA powder in 6 mL of 2:1 (v/v) acetone/*N*, *N*-dimethylacetamide at 30 °C. Then, 5 mL of the solution was loaded into a glass syringe equipped with a blunted stainless steel needle (diameter: ~0.9 mm) attached to a pump. Electrospinning was performed by ejecting the CA solution from the needle tip at a feeding rate of 0.4 mL h^{-1} toward the rotating drum, which functioned as a grounded collector under an electric field with an electrical potential of 17.5 kV from a high voltage supply. The distance between the needle tip and the aluminum foil (standard household quality with a thickness of about 15 μm) wrapped around the drum (circumference: approx. 19 cm) was 15 cm. The drum was set to rotate at 350 rpm, and the nanofibers were collected after 6 h. After electrospinning, the aluminum foil containing a nanofiber layer was removed from the collector. For the preparation of ABT-containing nanofibers, ABT powder was simply mixed with the CA solution to achieve the desired ABT concentration, and the mixed solution was electrospun under the optimized spinning conditions. Twelve nanofiber samples with the ABT concentration varying from 1.5 to 4.8 wt% (increment of 0.3 wt%) were prepared. SEM (scanning electron microscopy) images of two selected nanofiber samples and pictures of an actual sample are shown in the Supplementary data. The thicknesses of the samples were approximately 60 μm.

2.2. Acquisition of IR, Raman, and NIR spectra of the samples

To initially examine the distribution homogeneity of ABT in a sample, IR mapping was performed using transmission mode. An ABT/CA nanofiber sample was placed on a microscope stage connected with an IR spectrometer (LUMOS, Bruker), and IR radiation was focused to an area of 100 × 100 μm^2 for mapping. Each IR spectrum was collected with 32 scans (resolution: 4 cm^{-1}). Next, to acquire IR spectra of a sample area of 4 × 4 mm^2, the sampling compartment of the spectrometer (ABB Bomem) was modified to provide an illumination area of approximately 5 × 5 mm^2 using a concave lens. Then, an Al plate with a 4 × 4 mm^2 hole was positioned in front of a sample for measurement (128 scans). The resolution of the acquired spectra was 4 cm^{-1}.

NIR spectra were collected in transmission mode using an NIR spectrometer (ABB Bomem) equipped with a quartz-halogen light source and an InGaAs detector. An ABT/CA nanofiber sample was attached and fixed on a metal holder with a hole in the center. The resolution of the acquired spectra was 4 cm^{-1}. A Raman spectrometer (Kaiser Optical Systems) with a microscope stage was used for acquisition of Raman spectra. A nanofiber sample was positioned on the microscope stage, and a laser (785 nm) was irradiated onto the sample by an objective lens (10 × 0.25 numerical aperture (NA)) for

measurement. The laser spot size was approximately 50 μm in diameter. Each Raman spectrum (resolution: 4 cm^{-1}) was obtained by accumulating 50 scans with an exposure time of 3 s.

3. Results and discussion

3.1. Investigation of the IR spectral features of the ABT, CA, and 3.0 wt% ABT nanofiber samples

Fig. 1 shows the IR spectra of pure ABT (blue) and CA nanofiber (black) samples in the 1900–500 cm^{-1} range. A KBr pellet incorporating ABT was used to obtain the IR spectrum of the ABT sample, and the CA nanofiber sample was directly obtained in the transmission mode without further modification. In the IR spectrum of the CA fiber sample, major bands at 1740 cm^{-1} (C═O, carbonyl stretch of ester group), 1370 cm^{-1} (C–H, symmetric bending of methyl group), 1234 cm^{-1} (C–C–O, asymmetric stretching of carboxylate group), and 1049 cm^{-1} (C–O–C, asymmetric stretching in pyranose ring) were clearly observed [30,31]. For ABT, many narrow bands appeared. However, most of the bands, especially in the 1500–850 cm^{-1} range, were heavily overlapped with those of CA. Next, the spectrum of the 3.0 wt% ABT nanofiber sample (red) was obtained. The spectral features were similar to those of pure CA since the ABT concentration was only 3.0 wt%. Meanwhile, as shown in the insets, the ABT peaks at 1512 cm^{-1} and 777 cm^{-1} were apparent without serious overlap with the underlying CA feature, as expected. Therefore, the absorbances of these bands could be directly utilized to determine the ABT concentration. The 1512 cm^{-1} band with higher sensitivity (absorptivity) was solely used for further analysis.

One important issue to examine for accurate quantitative analysis is the distribution homogeneity of ABT over the nanofibers. If ABT is considerably localized, an acquired IR spectrum may not be representative of the entire ABT concentration in a nanofiber patch, and the accuracy subsequently degrades, especially when the sample coverage area is insufficient. To investigate the distribution homogeneity, a 3.0 wt% ABT sample was again selected, and IR mapping was performed on the sample. The mapping intervals were 100 and 100 μm in the x and y directions, respectively, and a total of 64 spectra was collected over an area of 800 × 800 μm^2. Fig. 2 shows the IR mapping image of the sample constructed using the peak area of the 1512 cm^{-1} band. The baselines of the spectra were initially corrected at 1518 and

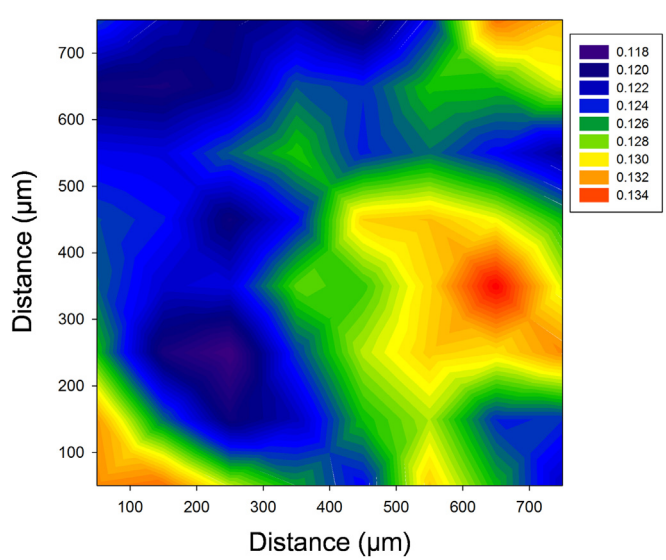

Fig. 2. IR mapping image of the 3.0 wt% ABT nanofiber sample constructed using the peak area of the 1512 cm^{-1} band. The mapping intervals are 100 μm in both the x and y directions.

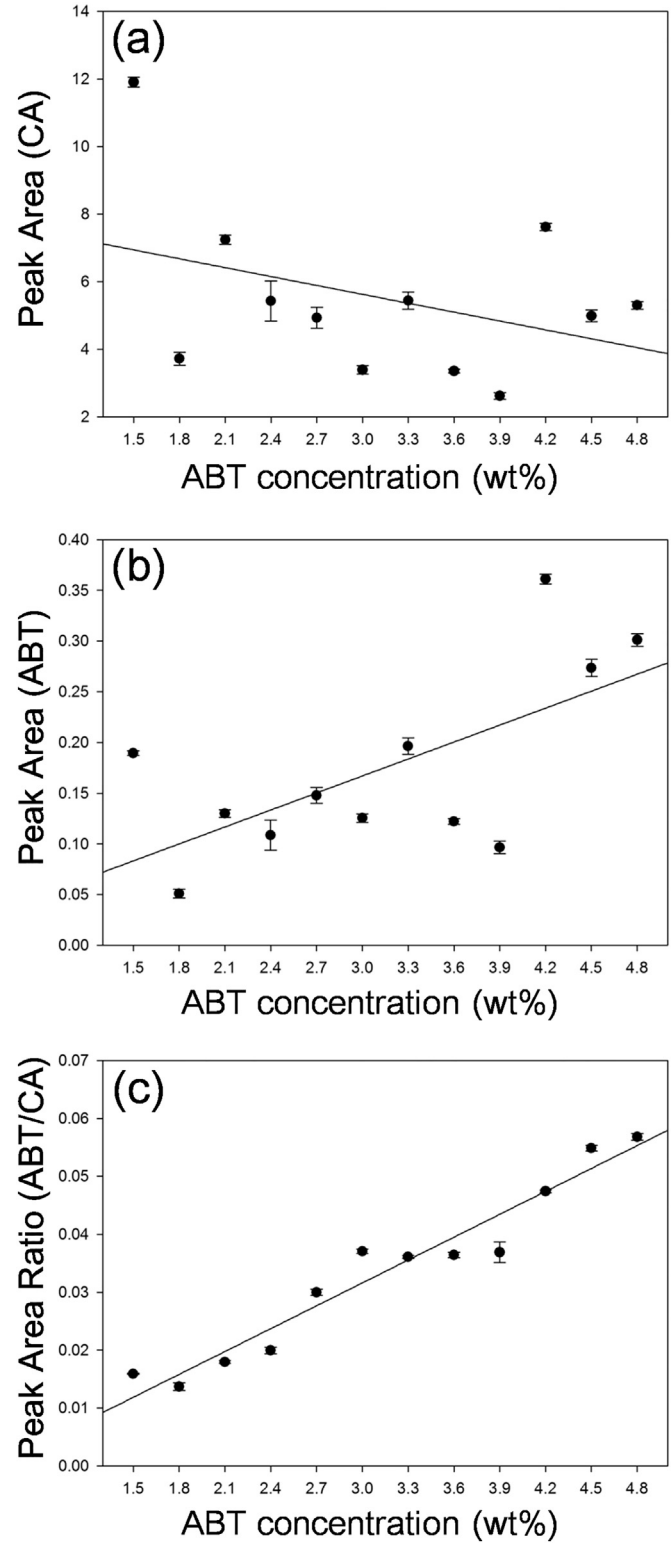

Fig. 3. Variations of (a) CA peak area, (b) ABT peak area, and (c) ABT/CA peak area ratios according to change of ABT concentration, where the IR sampling area was 800 × 800 μm^2.

1502 cm^{-1} for calculation of peak area. As observed, the color distribution (equivalent to the ABT distribution) was not perfectly uniform, and ABT was relatively more localized at the left-bottom. It is necessary to note that the magnitude of the peak intensity variation in the mapping area is not substantial, with a relative standard deviation (RSD) of 3.3%. Overall, the fluctuation of ABT concentration over the

mapping area was not significant, and microscopic heterogeneity in the distribution of ABT was observed.

3.2. Determination of ABT concentration in nanofiber samples

To determine ABT concentrations in the samples, average spectra acquired for each mapping (64 spectra) were used for the analysis. The peak areas of CA (1370 cm^{-1} band) and ABT in the 1402–1343 cm^{-1} and 1518–1502 cm^{-1} ranges, respectively, were calculated for each sample. Fig. 3(a) and (b) show the variations of the CA and ABT peak areas according to ABT concentration, respectively. For the CA peak, the area seemed to decrease with increasing ABT concentration. However, the trend is not clear. Also, the correlation between the ABT concentration and the corresponding peak area was very crude (Fig. 3(b)), with an R^2 value of 0.422. This poor correlation was attributed to the intensity fluctuation of the ABT peak due to the variation of sample thickness. For example, a larger amount of ABT might be loaded in a thicker nanofiber, which in turn results in an increase of ABT peak intensity.

One way to overcome the above-mentioned problem is the use of a peak ratio method to compensate for the peak intensity fluctuation due to the thickness variation of sample, since the intensities of ABT and CA peaks concurrently vary in parallel with the thickness. Therefore, the ratio between the intensities of these two peaks is expected to be thickness-independent. Fig. 3(c) shows the correlation between the ABT concentration and the peak area ratio. A linear relationship was clearly apparent, with an R^2 value of 0.941. However, the data was somewhat scattered around the regression line. The most plausible cause of the data scatter is that the spectra collected with the sampling area of $800 \times 800\,\mu m^2$ might not be sufficient to correctly represent the overall ABT concentration in a large nanofiber sample (actual size: 15 mm × 15 mm).

To collect more concentration-representative spectra of the samples, the illumination area of IR radiation was increased to $4 \times 4\,mm^2$, which is 25 times larger than that in the above measurement, and the acquired spectra were used for the analysis. Fig. 4(a) shows the correlation between the ABT concentration and the peak area ratio when the IR sampling area was $4 \times 4\,mm^2$. The data became much less scattered and closer to the regression line (R^2: 0.993). This clearly demonstrates that the coverage of a large area on the sample in spectral acquisition clearly helped to acquire IR spectra that are more representative of the concentration of each sample. To validate the independent prediction accuracy of the method, separately prepared ABT nanofiber samples (12 samples, concentration range from 1.5 to 4.8 wt%) were measured in the same configuration, and their ABT concentrations were determined using the linear regression model determined above. Fig. 4(b) shows the reference and predicted concentrations. As presented, the ABT concentrations were accurately predicted with a standard error of prediction (SEP) of 0.10 wt%. The sample coverage area of $4 \times 4\,mm^2$ clearly enabled an improvement of the accuracy in the determination of ABT concentration.

3.3. Evaluation of the Raman and NIR spectra of the 3.0 wt% ABT nanofiber sample

Raman and NIR spectroscopy are potential alternatives for analysis of the same samples since they provide analogous vibrational structural information of a sample through non-destructive measurements. Therefore, both the Raman and NIR spectra of the 3.0 wt% ABT nanofiber sample were collected, and the subsequent spectral features were examined to determine their applicability for analysis. Fig. 5 shows the Raman spectra of the 3.0 wt% ABT nanofiber, pure ABT, and CA nanofiber samples in the 1300–400 cm^{-1} range. Raman spectra of the 3.0 wt% ABT nanofiber and CA nanofiber samples are separately displayed at the bottom for detailed examination. As shown, the intensities of the CA nanofiber peaks were considerably low since the sample was

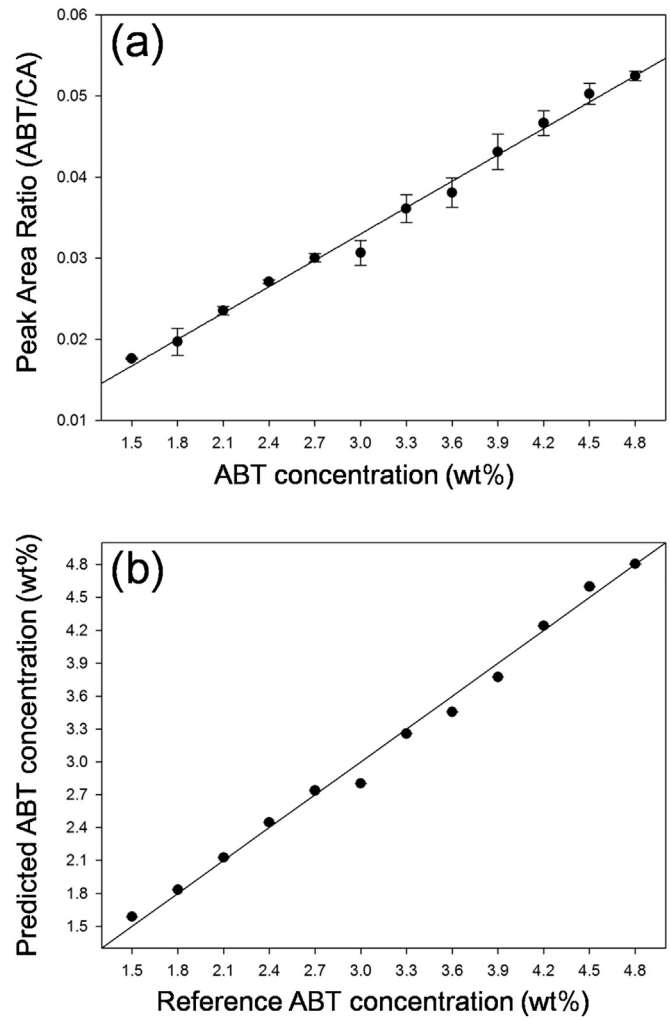

Fig. 4. (a) The correlation between ABT concentration and peak area ratio (ABT/CA) when the IR sampling area is $4 \times 4\,mm^2$ and (b) the predicted concentration (reference vs. predicted concentrations) for separately prepared samples using the linear regression model determined in the above plot.

very thin (~60 μm), equivalent to a small Raman cross-section. In the Raman spectrum of the 3.0 wt% ABT nanofiber sample, it was difficult to observe characteristic ABT peaks on the spectral features of CA nanofibers. Therefore, the sensitivity of the Raman measurements was not sufficient to accurately track the change of ABT concentration.

Fig. 6(a) shows the NIR spectra of the 3.0 wt% ABT nanofiber and pure CA nanofiber samples in the 8000–5000 cm^{-1} range when air was used as a background. In both cases, the baselines varied nonlinearly along the spectral range, and the absorption feature of ABT was difficult to recognize. The greater absorbance in the higher wavenumber (shorter wavelength) range did not originate from the larger absorption of NIR radiation by the nanofibers, but greater scattering resulted in attenuation of the radiation. As is generally known, the degree of light scattering becomes greater as the wavelength of radiation decreases. To compensate for the wavelength-dependent baseline variation, the NIR absorbance spectrum of the 3.0 wt% ABT nanofiber sample was produced again using pure CA nanofibers as the background (refer to Fig. 6(b)). Although the magnitude of the baseline variation was substantially reduced (the *y*-scale is much smaller), the nonlinear baseline was still present, and no noticeable spectral features related to ABT were observed. Therefore, the sensitivity of the NIR measurement was also insufficient for the analysis. As is known, NIR analysis is generally not sensitive since combination and overtone bands with low molar absorptivity are mostly present in the NIR region. In a meanwhile, a

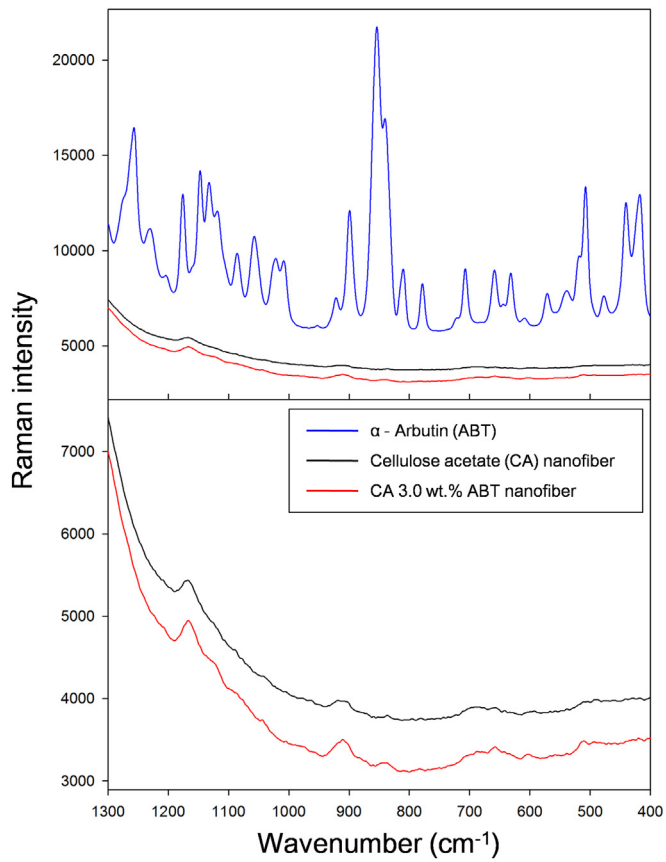

Fig. 5. Raman spectra of 3.0 wt% ABT nanofiber, pure ABT, and CA nanofiber samples in the 1300–400 cm^{-1} range. The Raman spectra of 3.0 wt% ABT nanofiber and CA nanofiber samples are separately displayed at the bottom for detailed examination.

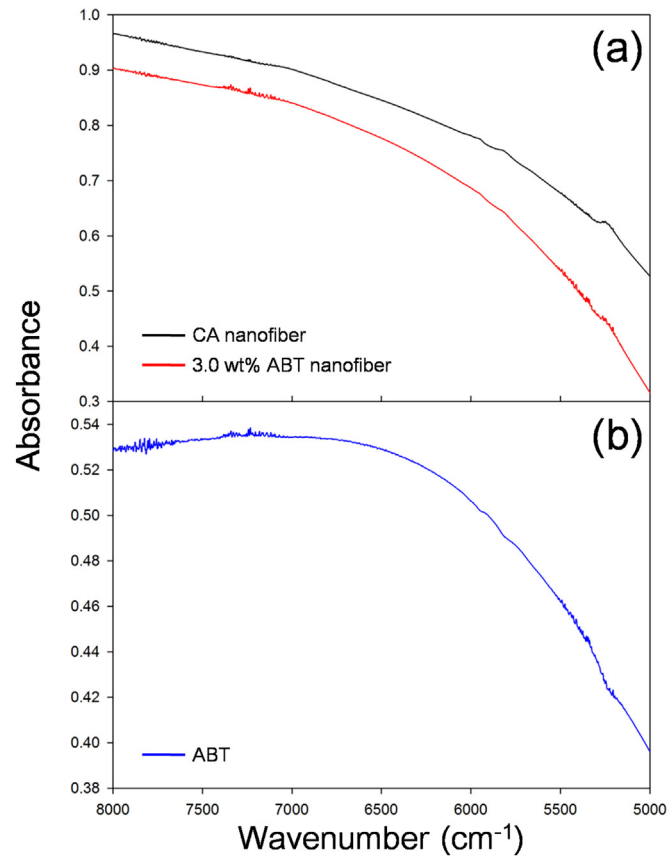

Fig. 6. (a) NIR spectra of the 3.0 wt% ABT nanofiber and pure CA nanofiber samples in the 8000–5000 cm^{-1} range, where air was used as a background. (b) NIR absorbance spectrum of the 3.0 wt% ABT nanofiber sample produced using CA nanofibers as a background.

nonlinear baseline was not observed in the IR measurement. Since the wavelengths of IR radiation are longer than those of NIR radiation, the magnitude of scattering is substantially smaller, and the wavelength-dependent baseline variation is thereby not significant.

4. Conclusion

The acquisition of IR spectra correctly representing the ABT concentration in a nanofiber matrix was the key for ensuring accuracy of the concentration determination where the IR sampling area was $4 \times 4\,mm^2$. Transmission IR spectroscopy can be an appropriate method for non-destructive determination of API concentration in diverse nanofiber-based patches. On the other hand, ATR (attenuated total reflection) IR measurements are not desirable since they require contact with a sample for spectral acquisition, leading to potential contamination. In addition, coverage of a wide area on a sample is practically limited due to the requirement of time-consuming point-to-point contact. In parallel, near-infrared (NIR) spectroscopy is more versatile when analyzing patches with greater thickness, since the transmission of IR radiation through these samples would not be sufficient to collect spectra with a good signal-to-noise ratio. When an API is strongly Raman-active, the adoption of Raman spectroscopy is recommended, but potential concurrent occurrence of fluorescence should be considered. Overall, the use of IR spectroscopy was appropriate in this study, while NIR and Raman spectroscopy are potentially more useful depending on the characteristics of nanofiber-based pharmaceutical patch samples. Nonetheless, the acquisition of concentration-representative spectra is desirable, regardless of the vibrational spectroscopic method used.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.microc.2018.04.008.

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